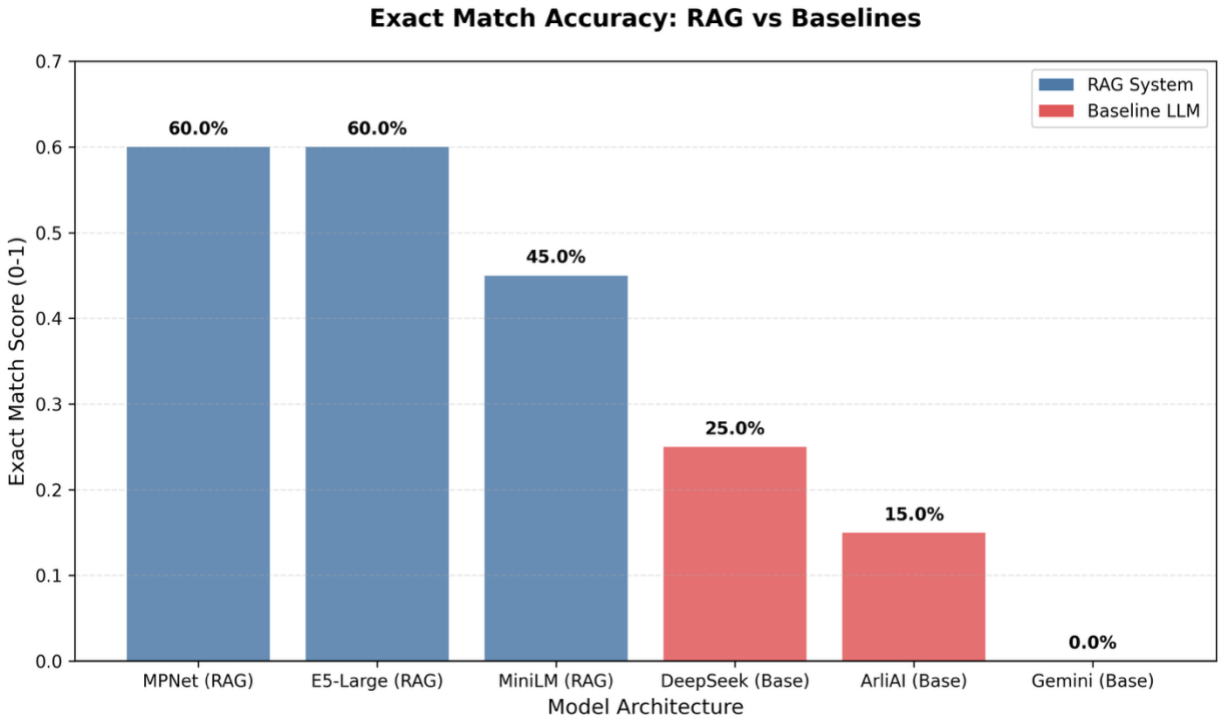
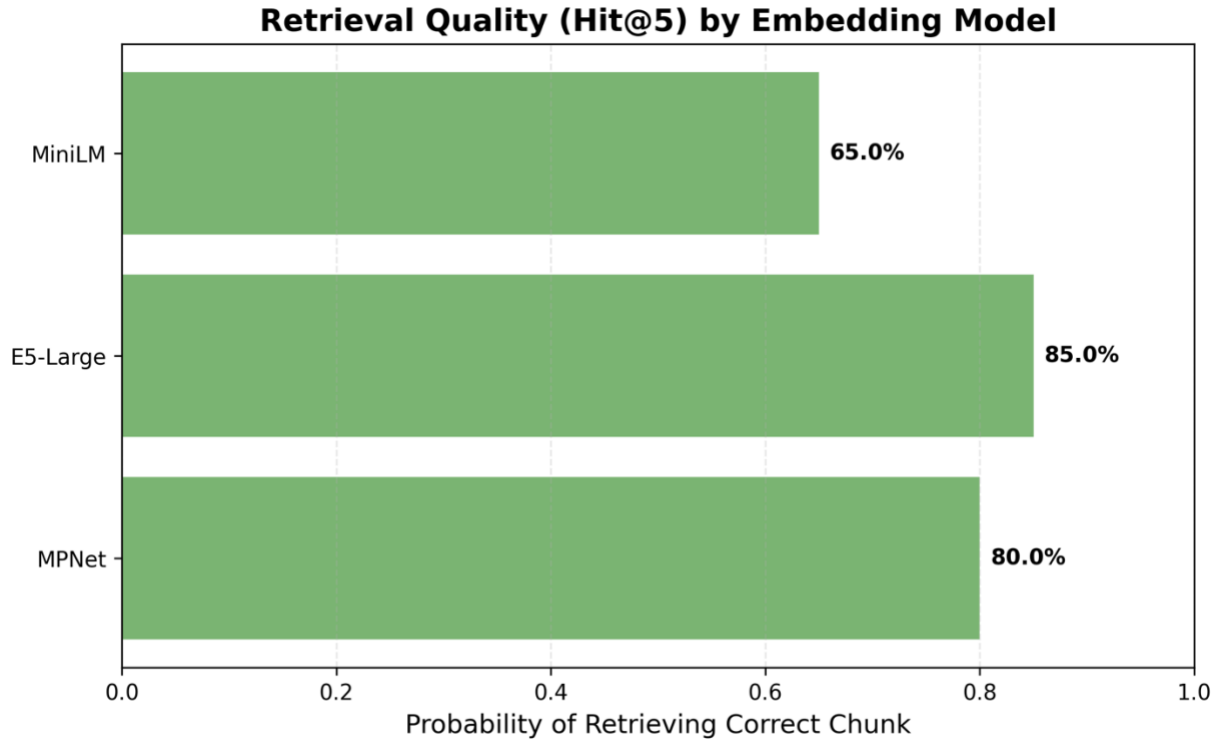
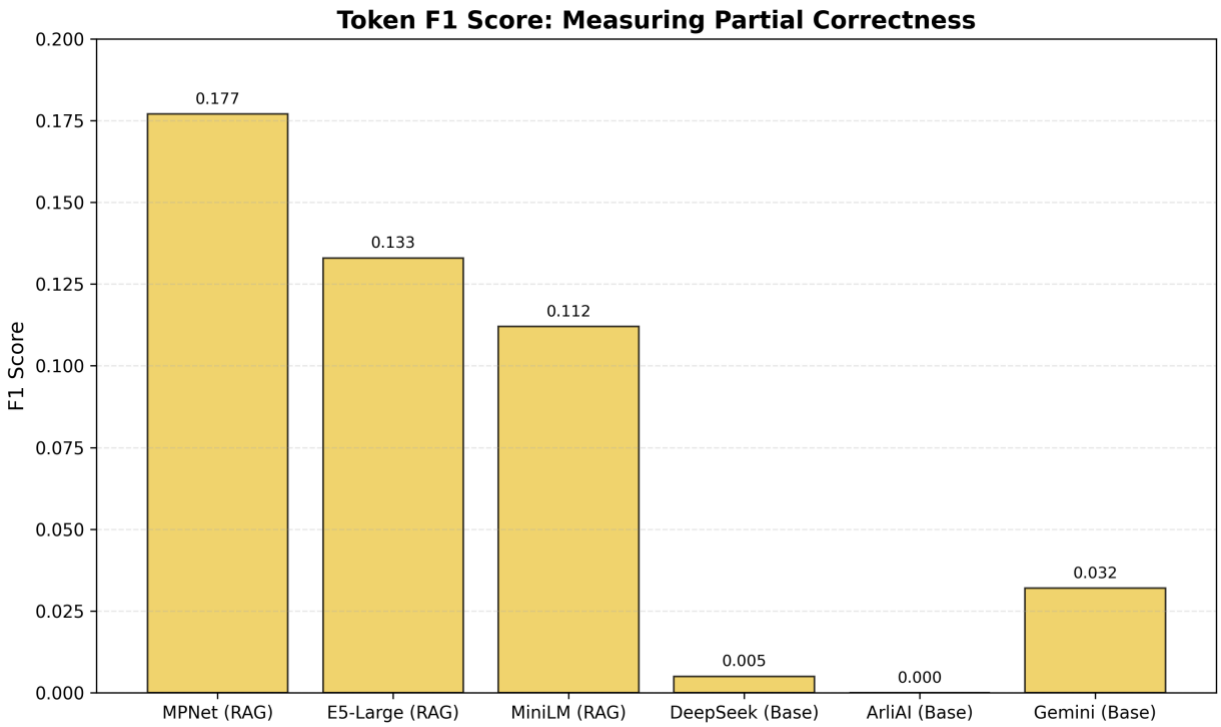
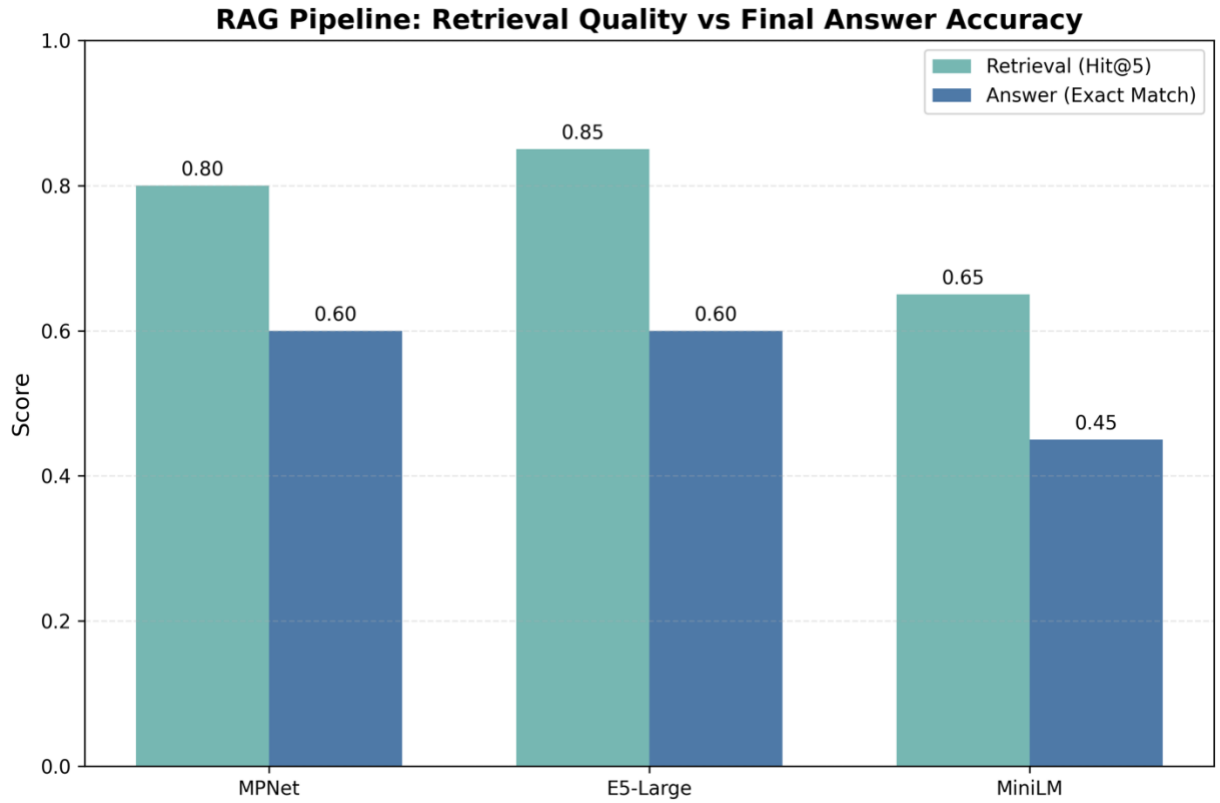
Appendix:

Visualizations :






Token F1 Score: Measuring Partial Correctness
F1 Score
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.177
0.133
0.112
0.005
0.000
0.032
MPNet (RAG)
E5-Large (RAG)
MiniLM (RAG)
DeepSeek (Base)
ArliAI (Base)
Gemini (Base)


RAG Pipeline: Retrieval Quality vs Final Answer Accuracy
Score
1.0
0.8
0.6
0.4
0.2
0.0
Retrieval (Hit@5)
Answer (Exact Match)
0.80
0.60
0.85
0.60
0.65
0.45
MPNet
E5-Large
MiniLM

Team Contributions:

- **Gaurang Agrawal (ga160):** Lead on Evaluation and Benchmarking. Developed the eval.py harness to rigorously test the system against baselines using Exact Match and Hit@K metrics.
- **Sawaiz Fatar (msf59):** Lead on Data Ingestion and Parsing. Implemented PyPDF2 logic to strip formatting noise and prepare raw text for the pipeline.
- **Skylar Qiu (zq67):** Lead on Chunking Strategy and Data Preprocessing. Designed the sliding window approach (600-token chunks) to optimize context retention.
- **Marwa Bouabid (mb951):** Lead on Modeling and Architecture. Selected and integrated the sentence-transformers library and evaluated embedding model performance.
- **Arshad Rizvi (ar845):**) Lead on Pipeline Integration. Engineered the core RAG logic in Untitled2.py, connecting the FAISS vector index with the LLM generation layer.

Instructions to run the code:

1) Prep a virtualenv

- From the project root (/Users/marwa/Desktop/final proj):
- python3 -m venv .venv
- source .venv/bin/activate
- Install deps:
- python3 -m pip install -r requirements.txt
- python3 -m pip install uvicorn google-generativeai

  (Optional: export TOKENIZERS_PARALLELISM=false to silence tokenizer fork warnings.)

2) Required API keys (export in each shell before running servers/evals)

- Baseline (DeepSeek/OpenAI-compatible):
- export OPENAI_API_KEY=" sk-da3a48a8cc914ab9b46f6c3716129ffc "
- export OPENAI_BASE_URL="https://api.deepseek.com"
- Gemini:
- export GOOGLE_API_KEY="

  AIzaSyBKeTbzX2Cy7nMlNO_YFLbvXM2sUtoOEoc "
- ArliAI:
- export ARLIAI_API_KEY="790aca55-e6dc-4fd0-b173-c1ad32b48c74"
-

3) Run the backend (chatbot)

- In an active venv with the above keys set (at least OPENAI_API_KEY for your RAG pipeline):
- uvicorn server:app --reload --port 8000
- Use the app:
  - Open index.html directly (double-click or open index.html), or
  - Visit http://localhost:8000/ (served by FastAPI).

4) Evaluate RAG/baselines/Gemini/ArliAI

- With venv active and keys exported, run the automation script:
- bash run_all.sh

  This will:

  - Sweep RAG over 6 embeddings (rebuilds FAISS each time): all-mpnet-base-v2, all-MiniLM-L6-v2, intfloat/e5-base-v2, intfloat/e5-large-v2, multi-qa-MiniLM-L6-cos-v1, gtr-t5-base.
  - Run baseline once (deepseek-chat).
  - Run Gemini once (gemini-1.5-flash).
  - Run ArliAI once (Gemma-3-27B-it).
- Results are written alongside eval_qa.jsonl with suffixes like .rag.results.jsonl, .baseline.results.jsonl, .gemini.results.jsonl, .arliai-<model>.results.jsonl.

5) If ArliAI/Gemini calls hang

- Test connectivity with curl first (adjust model/key as needed):
- curl -i https://api.arliai.com/v1/chat/completions \
- -H "Authorization: Bearer $ARLIAI_API_KEY" \
- -H "Content-Type: application/json" \
- -d '{"model":"Gemma-3-27B-it","messages":[{"role":"user","content":"hi"}],"max_tokens":32}'
- curl -i https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?key=$GOOGLE_API_KEY \
- -H 'Content-Type: application/json' \
- -d '{"contents":[{"parts":[{"text":"hi"}]}]}'

6) Starting fresh

- If you hit embedding index issues, delete faiss.index and index_meta.json, then rerun RAG (the script does this per embedding).

7) Quick manual eval runs

- Single RAG run with current embedding: python3 eval.py --qa-file eval_qa.jsonl --mode rag --k 5

- Baseline: `python3 eval.py --qa-file eval_qa.jsonl --mode baseline --baseline-model deepseek-chat`
- Gemini: `python3 eval.py --qa-file eval_qa.jsonl --mode gemini --gemini-model gemini-1.5-flash`
- ArliAI: `python3 eval.py --qa-file eval_qa.jsonl --mode arliai --arliai-models "Gemma-3-27B-it" --arliai-base-url https://api.arliai.com/v1 --arliai-timeout 15`
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